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Review

Recent progress in understanding starch gelatinization - An important property determining food quality

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ABSTRACT

Starch gelatinization is a crucial process in determining both texture and nutrition properties of starch-based foods, while its complex nature is still not fully understood. Kinetics modeling has been recently developed for understanding starch gelatinization under both limited and excessive water content. Amylose with different chain lengths has distinct effects on starch gelatinization temperatures and enthalpy by interacting with amylopectin chains in semi-crystalline lamella. Moisture is a crucial factor in determining starch gelatinization property, with new evidence suggesting that different groups of amylopectin-amylopectin or amylose-amylopectin double helices are involved for multi-endothermic gelatinization peaks under limited moisture content. The presence of salts, sugars, protein, lipids and non-starch polysaccharides can affect starch gelatinization through various mechanisms. All these new insights and future directions in terms of better understanding starch gelatinization property were summarized. This information could help develop new generations of foods with desirable properties through a better understanding of starch gelatinization process.

1. Introduction

Starch in foods such as rice, noodle and breads is frequently gelatinized before human consumption. During gelatinization, starch undergoes an order to disorder phase transition, which can irreversibly disrupt starch semi-crystalline structures and largely determine textural and nutritional properties of starch-based foods (Chung, Lim, & Lim, 2006; Fariña, Torres, & Moreira, 2019; Martinez, Roman, & Gomez, 2018; Li, Gong, Huang, & Yu, 2021; Li, Hu, Gu, & Gong, 2021). Limited degree of starch gelatinization in various foods could be an efficient way to slow down starch digestibility and generally improve human health in terms of maintaining a stable postprandial glycemic response (Chung et al., 2006; Wang & Copeland, 2013). This is typically important considering the global pandemic of chronic diseases including obesity and type 2 diabetes. On the other hand, starch gelatinization degree is negatively correlated with the hardness and sensory perception of many foods such as baked foods and cooked rice (Lin, Pan, Liu, & Wei, 2019; Martinez et al., 2018). A better understanding and manipulation of starch gelatinization property is thus highly needed in terms of developing healthier foods (i.e. with slow starch digestibility) with ideal textural attributes.

Many new insights have been learnt recently with respects to the nature/ structural basis for starch gelatinization property (Fig. 1) (Li, Cao, Cao, & Li, 2022; Niu, Zheng, Fu, Zeng, & Liu, 2022; Sun et al., 2021). For example, the relative length of amylose intermediate chains has been found to be negatively correlated with the gelatinization temperature range of rice starches; onset and peak gelatinization temperature are negatively correlated with the relative length of amylopectin short chains, while the conclusion temperature is related to the relative length of amylopectin medium chains (Li & Gong, 2020). In addition, new insights on the effects of moisture content and primary food ingredients such as salts, sugars, protein, lipids and non-starch polysaccharides on starch gelatinization property have also been learnt (Fig. 1) (Li et al., 2022; Nicol, Isobe, Clark, Matubayasi, & Shimizu, 2019; Renzetti, Hoek, & Sman, 2021), although there are also many unknowns remaining to be further studied in the future.

This review aimed to summarize all these advances, as well as the unknowns and future directions in terms of better understanding starch gelatinization property (Fig. 1). Considering the importance of starch gelatinization in determining the food quality, this review contains important information that can potentially help the development of next generations of healthier foods with desirable textures and nutrition.

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2. Characterization and mathematical modeling

Starch gelatinization involves multiple transitions, which can be generally described by the framework designed for synthetic side-chain liquid crystalline polymers (Waigh, Gidley, Komanshek, & Donald, 2000). During the initiation of starch gelatinization, water enters the amorphous growth rings, where starch granules start to swell and amylose molecules start to leach out (Jenkins & Donald, 1998; Wani et al., 2012). Only when amorphous background significantly swelled, sufficient stress could then be imposed through connecting molecules from amorphous to crystalline regions, initiating the disruption of the crystallites. During this stage, helix-helix side chains firstly dissociate followed by helix-coil transition (unwinding of double helices) (Waigh et al., 2000).

Different techniques have been applied to characterize these structural transitions during starch gelatinization, including differential scanning calorimetry (DSC), rapid visco-analyzer (RVA), rheometric measurement, solid-state nuclear magnetic resonance (NMR), optical/polarized microscope, small-angle X-ray scattering and X-ray diffractometry (Ahmed, 2012; Niu et al., 2022; Waigh et al., 2000; Wang & Copeland, 2013; Wani et al., 2012). Each technique has its own advances and limitations, which have been extensively summarized in previous reviews (Schirmer, Jekle, & Becker, 2015; Wang & Copeland, 2013). For examples, although optical microscopy has been widely used to investigate the gelatinization behavior of starch granules, the measurement of gelatinization temperatures is not quite accurate and the energy absorbed during gelatinization cannot be determined. Therefore, for a quantitative measurement, DSC, which can measure the heat absorbed or released by a starch sample during phase transitions, is considered as the technique for the precise determination of gelatinization temperatures and enthalpy (Wang & Copeland, 2013).

DSC under non-isothermal conditions can produce a thermogram of endothermic heat flow against time. A series of empirical parameters, including gelatinization onset (To), peak (Tp) and conclusion (Tc) temperature, as well as endothermic enthalpy (ΔH) and peak height index can be then obtained. These parameters are generally believed to be able to reflect the ease and energy required to melt starch crystallites (i.e. ordering and amount of starch double helices) (Wang & Copeland, 2013; Zanoni, Schiraldi, & Simonetta, 1995). For example, starch crystallites with least stability are believed to melt at low temperature, contributing to the To temperature; the ones with a higher quality are then melted at a much higher temperature, contributing to the Tc temperature; gelatinization temperature range could thus indicate the homogeneity of stash crystallites (Sandhu & Singh, 2007). Gelatinization enthalpy (ΔH) is correlated with the ordering and amount of starch crystallites (Wang & Copeland, 2013). Therefore, these DSC parameters have been frequently applied to investigate starch structure-gelatinization relations in the literature (e.g. (Li et al., 2022; Li et al., 2020).

However, it has been recently shown that Tc might not always reflect the completion of gelatinization for all starches. For instance, Tc indicates the completion of gelatinization for rice starch, while residual structural order can be found in wheat and maize starch (Wang et al., 2018). Instead, maize and wheat starch reached the completion of their gelatinization at a temperature higher than Tc, which was defined as the end temperature (Te) (Wang et al., 2018). In addition, heating starch-water mixture at a temperature below To has resulted in disruption of both short- and long-range ordered starch structure, indicating that starch gelatinization initiated earlier than To (Liu, Yu, Copeland, Wang, & Wang, 2019). A careful interpretation of traditional DSC

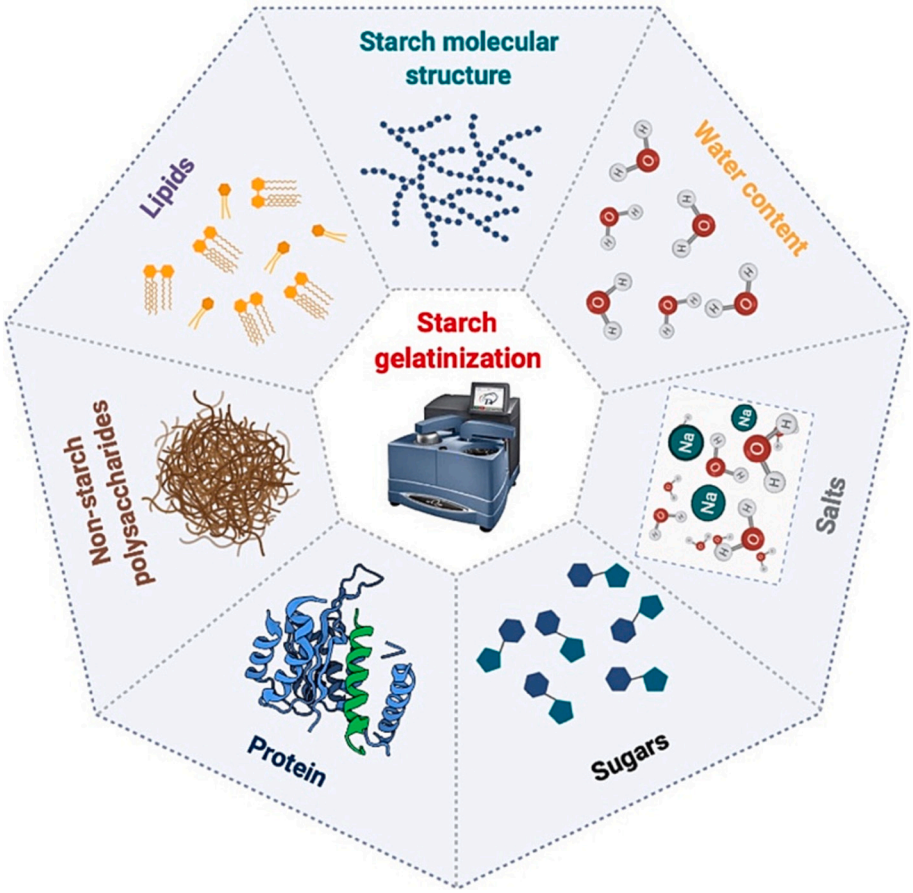

Fig. 1. Factors that are important for determining starch gelatinization property, including starch fine molecular structures, water content, presence of salts, sugars, protein, lipids and non-starch polysaccharides.

gelatinization parameters is thus required in terms of representing the actual starch structural change during the gelatinization process. A new parameterization method to produce gelatinization parameters might be urged, to be better related to starch structures.

Furthermore, traditional DSC parameters may not be able to capture some fine features in the DSC thermograms (i.e. number and positions of peaks), especially when limited water content is applied (Li & Hu, 2021b). Different mathematical models have thus been developed to parameterize DSC endotherms and better understand starch gelatinization property. Generally, starch gelatinization in excessive water follows the first-order kinetics under isothermal condition (Table 1), with its rate constant depending on temperature, which can be described according to the Arrhenius equation ($k = k_0 e^{-\frac{E_a}{RT}}$, where k_0 is the pre-exponential factor; E_a is the activation energy; T is the absolute temperature and R is the universal gas constant) (Turhan & Gunasekaran, 2002; Zanoni et al., 1995). That is, the formation rate of gelatinized starch mass is proportional to the ungelatinized starch amount. It has been shown that gelatinization rate constant was in the order of 10^{-4} to 10^{-3} s^{-1} for the in vitro gelatinization and 10^{-5} to 10^{-4} s^{-1} for the in situ gelatinization of wheat starch; the average activation energy (E_a, its value is related to the energy barrier that has to be overcome for gelatinization to occur) was about 76 kJ/mol for wheat starch when gelatinized between 60 °C and 100 °C (Turhan & Gunasekaran, 2002). Note, starch gelatinization only follows first-order kinetics beyond the gelatinization onset temperature, as gelatinization only affect the amorphous regions during the initial heating phase (Zanoni et al., 1995). n^{th} order kinetics (i.e. $d\alpha/dt = k(1\text{-}\alpha)^n$, where α is the degree of starch gelatinization, n is the reaction order, k is the rate constant and $d\alpha/dt$ is the derivative of α with respect to time) has also been applied when starch gelatinized under non-isothermal conditions (Table 1) (Ahmed, 2012; Resio & Suarez, 2001; Spigno & Faveri, 2004; Zhang et al., 2013). The reaction order was very complex, which was related to the starch varieties, presence of sugar and salt, heating rate, as well as ranges of heating temperature (Ahmed, 2012; Zhang et al., 2013).

Most of the above studies have applied an excessive water content during starch gelatinization, which can produce a symmetric DSC endotherm. However, during the industrial processing for many starch-based foods (such as baked foods (Martinez et al., 2018)), multiple asymmetric peaks have been frequently observed from the DSC thermograms, due to the limited gelatinization moisture content (e.g. ≤ 60 %) (Liu, Yu, Xie, & Chen, 2006). High amylose starch and other modified starches (e.g. acid hydrolyzed and heat-moisture treated maize starch) can also often display a very broad gelatinization peak with different bumps even under adequate water (Li et al., 2019; Liu et al., 2006; Xing, Li, Wang, & Adhikari, 2017). To be able to consider and capture those fine features in DSC thermograms, a few mathematical models have been developed in the literature. For example, an empirical model applying the Gaussian function was applied to deconvolute the overlapped DSC endothermic peaks (Liu et al., 2006). Other mathematical kinetics-based models have also been developed (Ahmed, 2012; Zhang et al., 2013). The issues with these models are that their resulted parameters are either purely empirical and mathematical in nature (no physical basis), or there was no information with respects to if these models can satisfactorily reproduce the whole DSC thermogram. Therefore, a kinetics-based mathematical model has been recently developed to fit the asymmetric multi-phase gelatinization endotherm (Table 1), which could reproduce a parameterization of DSC thermogram that is essentially the same as the experimental one (Li, Hu, 2021b). The model has reduced the whole starch gelatinization *endo*-thermogram into a few kinetics-based parameters, including E_a (activation energy), n (reaction order), and $\Delta H\infty$ (maximum gelatinization enthalpy). A larger E_a value means a higher activation energy needed for the gelatinization reaction to occur, while higher n values indicate a faster reaction, which could possibly result in lower gelatinization temperatures. By reducing the whole DSC endotherm into a few kinetics-based parameters, the model enables correlation analysis between starch gelatinization property and starch structural parameters, which can help better understand the starch structure-gelatinization relations. For example, by applying this model to 9 different starches and correlating obtained DSC fitting parameters with the corresponding lamellar structures, distinctly arranged sub-components of semi-crystalline lamellae with distinct thermal stabilities have been identified in these starch granules (Li, Hu, 2021b).

3. Relations between starch molecular structure and its gelatinization property

Starch molecular structures are critical intrinsic factors that can largely determine starch gelatinization properties, which have been frequently observed in the literature (e.g. (Li & Gong, 2020; Li et al., 2020). According to amylopectin cluster model, amylopectin chains can be defined as A, B and C chains, with A (~ DP 6–12) and B1 (~ DP 13–24) chains forming double helices in the semi-crystalline lamella of starch granules (Hanashiro, Abe, & Hizukuri, 1996). Therefore, longer amylopectin double helices are normally associated with longer amylopectin A and B1 chains, which require a higher temperature to gelatinize (Kong, Bertoft, Bao, & Corke, 2008; Tao, Li, Yu, Gilbert, & Li, 2019). Amylopectin external A chains can be further divided into A_{fp} (~ DP 6–8) and A_{cltr} (~ DP 9–12) chains; A_{fp} are too short to form double helices, while A_{cltr} chains are preferred in terms of developing double helices (Zhu, 2018). The amount of A_{cltr} chains was found to be positively correlated with starch gelatinization temperatures (Nakamura et al., 2002; Noda et al., 2009; Zhu, Corke, & Bertoft, 2011). Gelatinization conclusion temperature was related to amylopectin double helices formed between trans-lamella chains (DP 37–69), which might form long amylopectin double helices protruding from crystalline lamellae intro continuing amorphous lamellae (Li & Gong, 2020). These trans-lamellar double helices might be more thermally stable and melt at the end of gelatinization process. In addition, longer amylopectin internal chain segments (i.e. chain segments between branching points)

Table 1
Different mathematical models for fitting starch gelatinization thermograms.

Kinetics model	Equation	Meanings of parameters	Suitable conditions	Reference
First-order kinetics	$\Delta H(t) = (\Delta H\infty) \times (1 - e^{-kt})$	$\Delta H(t)$ is the gelatinization enthalpy at time t; $\Delta H\infty$ is the maximum gelatinization enthalpy; k is the rate constant	Isothermal gelatinization with an excess of water	(Turhan & Gunasekaran, 2002; Zanoni et al., 1995)
n^{th} order kinetics	$\Delta H(t) = \Delta H\infty - \left[\Delta H\infty^{(1-n)} - (1-n)kt\right]^{\frac{1}{1-n}}$	$\Delta H(t)$ is the gelatinization enthalpy at time t; $\Delta H\infty$ is the maximum gelatinization enthalpy; k is the rate constant; n is the reaction order	Non-isothermal gelatinization with an excess of water	(Ahmed, 2012; Zhang, Tong, Zhu, & Ren, 2013)
Asymmetric multi-phase gelatinization kinetics	$\Delta H(t) = \sum_{i=1}^{\infty} \Delta Hi(t)$	$\Delta H(t)$ is the gelatinization enthalpy at time t; $\Delta Hi(t)$ is the gelatinization enthalpy of ith component at time t, which can be either described by a first-order or n^{th} order kinetics equation	Non-isothermal gelatinization with a limited moisture content	(Li, Hu, 2021b)

Note: the first-order and n^{th} order kinetics equations were rewritten from the original publications for a comparable interpretation with the newly developed asymmetric multi-phase gelatinization kinetics equation.

allow a more ordered alignment of amylopectin double helices in the crystalline lamella, which are more thermally stable (Kong et al., 2008). For example, the inter-block chain length was positively correlated with the gelatinization onset temperature of different starches (Vamadevan, Bertoft, & Seetharaman, 2013).

In terms of amylose molecular structures, amylose content is generally believed to be negatively correlated with starch gelatinization enthalpy, as an increase of amylose content would decrease the amount of amylopectin double helices (Li et al., 2019; Zhang et al., 2016). However, a recent study applying a series of rice starches with a wide range of amylose content (~ from 11 % to 26 %) showed that relations between amylose content and gelatinization enthalpy were much more complex, as it was neither parabolic or linear (Li et al., 2022). The contradiction between these findings was possibly due to the application of starches with a limited range of amylose content in earlier studies. Except for amylose content, other amylose fine molecular structures could also significantly determine the starch gelatinization property, as starches with a similar amylose content can often display a significantly different gelatinization property (Li, Hu, Tao, Gong, & Yu, 2020). It has been proposed that relative longer amylose intermediate chains could interact with amylopectin in the crystalline lamella by forming double helices, which can alter the homogeneity of amylopectin double helices and increase the gelatinization temperature range (Tc-To) (Li & Gong, 2020; Li & Gong, 2021). The interactions between amylose and amylopectin molecules into double helices were supported by earlier studies, which showed that the overall amount of starch double helices was much higher than the starch crystallinity determined by the long-range order of amylopectin double helices (Cooke & Gidley, 1992).

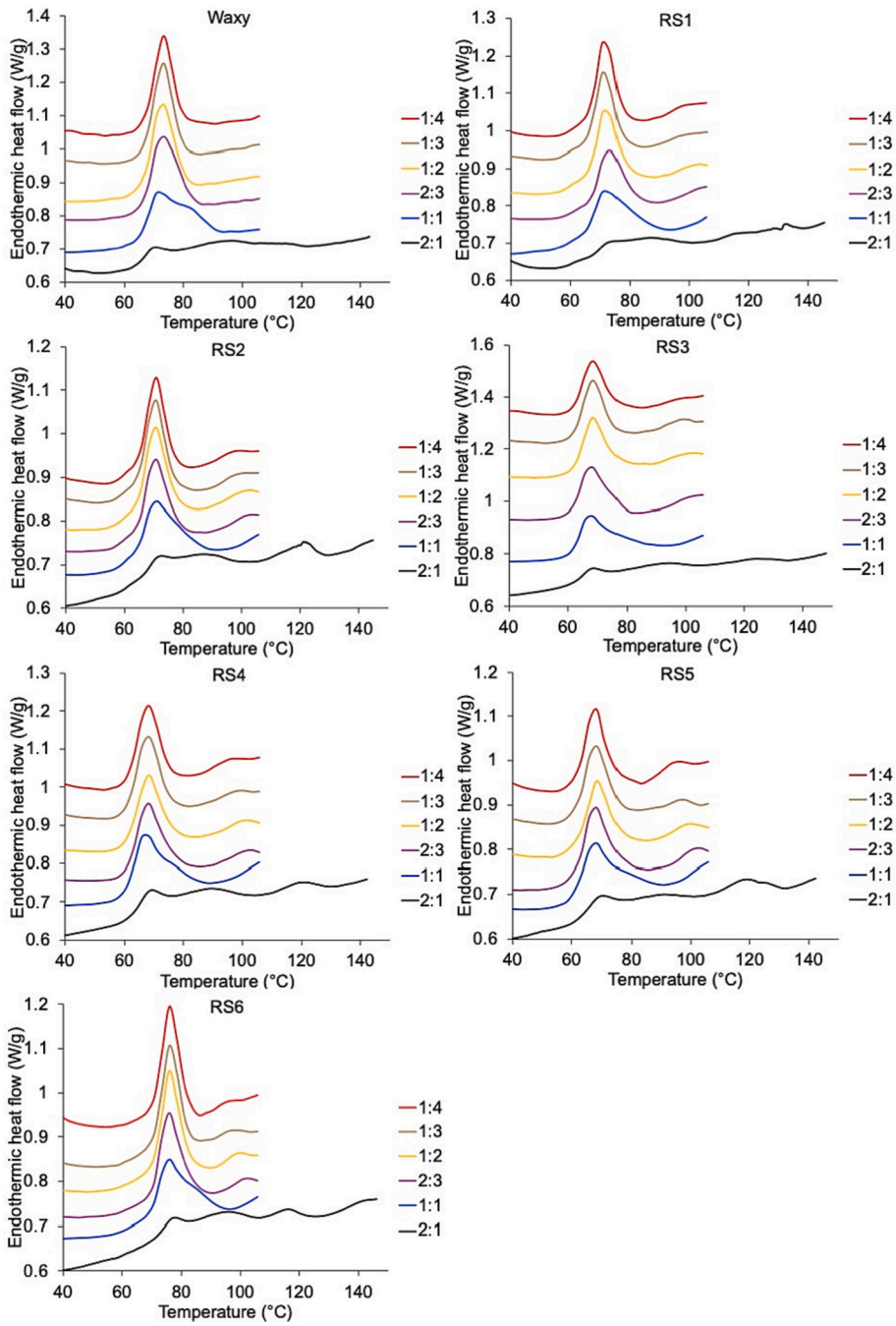

Fig. 2. DSC gelatinization thermograms for waxy maize and 6 different rice starches (RS1–6) at the starch to water weight ratio of 2:1, 1:1, 2:3, 1:2, 1:3 and 1:4. This graph is collected from the literature with permission (Li et al., 2022).

These amylose-amylopectin double helices had an intermediate ordering compared to the short/ unstable (controlled by To) and long/ stable (controlled by Tc) amylopectin double helices, as they seems do not have an impact on the gelatinization temperatures (i.e. To, Tp and Tc) (Li & Gong, 2020). Granule-bound starch synthase isoform I (GBSSI) has been proposed to be immobilized within amylopectin crystalline lamella, utilizing malto-oligosaccharide or amylopectin branches as primers to synthesize amylose molecules (Ball, van de Wal, & Visser, 1998). There is thus the possibility from the biological perspective that amylose can interact with amylopectin within the crystalline lamella during its biosynthesis considering the strong tendency of forming hydrogen bonds between hydroxyl groups. It was proposed that the interactions between amylopectin and amylose as double helices were not favored within amorphous lamella, as amylopectin internal chain length is too short (about DP 5) (Bertoft et al., 2016). However, amylose short chains can possibly form entanglements with amylopectin internal chains in the amorphous regions and further affect starch gelatinization property (Li & Hu, 2021a; Zhong et al., 2020).

4. Effects of moisture on starch gelatinization

Moisture content is one of the important gelatinization conditions that can largely affect starch gelatinization property (Wang & Copeland, 2013). Generally, a single and symmetric gelatinization peak is observed when the water content is in an excessive amount (e.g. above 65 %) (Donovan, 1979; Wang & Copeland, 2012). However, multiple gelatinization peaks could be observed from the DSC thermograms when limited water is presented (e.g. between 35 % and 60 %) (Li et al., 2022; Li, Hu, 2021b). For example, two gelatinization endothermic peaks were observed around 75 °C and 90 °C for rice starches with limited moisture content (Fig. 2) (Li et al., 2022). Another endothermic gelatinization peak around 120 °C was also observed for those normal rice starches gelatinized under limited moisture content, which was however not clear in the waxy maize starch, indicating that this gelatinization endothermic peak was related to amylose such as amylose-lipid complex and amylose-amylose interactions (Li et al., 2022; Liu et al., 2006). Furthermore, only a single endotherm might be observed when the amount of water decreased below 35 % during starch gelatinization (Donovan, 1979).

The structural basis for these multiphasic peaks are a complex issue to solve (Li, Hu, 2021b; Xing et al., 2017). Three hypotheses are available in the literature based on experimental observations in order to rationalize the appearance of different gelatinization peaks. The first hypothesis was proposed by Donovan at the year of 1979, which suggested that the DSC endothermic peak with lower temperatures was contributed by the swelling-driven disruption of starch crystallites, and endotherm with higher temperatures was derived from the melting of remaining less hydrated crystallites (Donovan, 1979; Waigh et al., 2000). The second hypothesis was proposed three year after, which indicated that starch granules contained crystallites with different thermal stabilities, with less stable crystallites melted first with sufficient water and more stable starch crystallites melted afterwards with limited water or in the absence of water (Evans & Haisman, 1982; Li, Hu, 2021b). The last hypothesis was proposed at the year of 1986 by Biliaderis et al., (Biliaderis, Page, Maurice, & Juliano, 1986). It suggested that an annealing process possibly involving the rearrangement of amylose or amylopectin double helices occurred during the gelatinization process, which can form an endothermic bump with higher gelatinization temperatures. Although some other studies have been conducted in order to elucidate the nature of these multiphasic gelatinization peaks afterwards, their findings can be generally classified according to these three hypotheses from the early time (Wang & Copeland, 2013).

However, although these hypotheses were all reasonable considering their experimental evidence, none of them have further investigated the molecular structural basis such as starch chain-length distribution and molecular size distribution giving rise to these gelatinization peaks. Combined effects of starch molecular structures and moisture content on starch gelatinization property has only been recently investigated (Li et al., 2022). It showed that amylose content had a parabolic relation with rice starch gelatinization temperatures. That is, rice starch with either a high (> 25 %) or low (< 15 %) amylose content has a relatively higher gelatinization temperature compared to starches with an intermediate amylose content (15 % - 25 %). Therefore, amylose may have a bi-directional effect on the stability of starch double helices, that either a high or low amylose content could facilitate the development of more ordered starch crystallites. Merely studying starches with a narrow range of amylose content could thus miss critical information with respects to relations between starch structure and gelatinization property. Two mechanisms were further proposed in the above-mentioned reference to rationalize the relations between amylose content and rice starch gelatinization temperatures: (1) higher gelatinization temperatures of rice starches were related to lower amylose content, as amylose molecules can interfere the perfect alignment of amylopectin double helices in crystalline lamella (Li & Gong, 2021; Li, Hu, 2021a); (2) higher gelatinization temperatures of rice starches were related to higher amylose content, as a higher amylose content is associated with longer amylopectin single lamellar chains and thus thicker crystalline lamella (Li et al., 2019).

More importantly, amylose short chains were suggested to be involved in the development of first endothermic gelatinization peak, and interactions between relatively shorter amylopectin trans-lamellar chains and amylose intermediate chains dominantly contributed to the formation of second gelatinization endothermic peak when starch was gelatinized under limited moisture content (Fig. 3) (Li et al., 2022). Starch has a complex and heterogenous semi-crystalline lamellae structure, consisting of not only amylopectin double helices but also double helices formed between amylopectin and amylose intermediate chains as well as entanglements of amylose short chains with amylopectin inter-block chains in the amorphous regions (Li & Gong, 2020; Li & Gong, 2021; Zhong et al., 2020). A single symmetric gelatinization endothermic peak is produced by melting these heterogenous double helices and entanglements with excessive moisture content. However, amylopectin double helices with an easy access to free water and weak entanglements between amylopectin inter-block chains and amylose short chains could be firstly gelatinized and contributed to the first endothermic gelatinization peak under limited moisture content, due to their less stability (Li et al., 2022). Rest amylopectin double helices and amylose-amylopectin double helices in the crystalline regions melt at higher temperatures and contribute to the formation of the second gelatinization peak due to the lack of free water and their improved quality of double helices (Li et al., 2022; Li & Gong, 2020).

5. Effects of salts on starch gelatinization

Starch is frequently gelatinized with the presence of many other food ingredients, such as salts, sugars, protein, lipids and non-starch polysaccharides. For example, both starch and salt are indispensable ingredients for yellow alkaline noodles (Yeoh, Lubowa, Tan, Murad, & Mat Easa, 2020). These ingredients can all have a significant influence on starch gelatinization (Donmez, Pinho, Patel, Desam, & Campanella, 2021). Therefore, understanding the effects of different food ingredients on starch gelatinization property could better help the development of food products with desirable properties.

The effects of addition of salts on starch gelatinization property could be very complex (Table 2). Depending on the types and concentrations, salts can cause either a depression or elevation of starch gelatinization temperature and enthalpy (Chiotelli, Pilosio, & Le Meste, 2002; Donmez et al., 2021). For example, the gelatinization peak temperature of indica, japonica and waxy rice starch was increased by 10–13 °C and enthalpy increased by 2–3 mw/mg when mixed with up to 0.6 mol/L K_2SO_4 (Yang, Xu, & Wang, 2021). When 0.1 mol/L NaCL solution was mixed

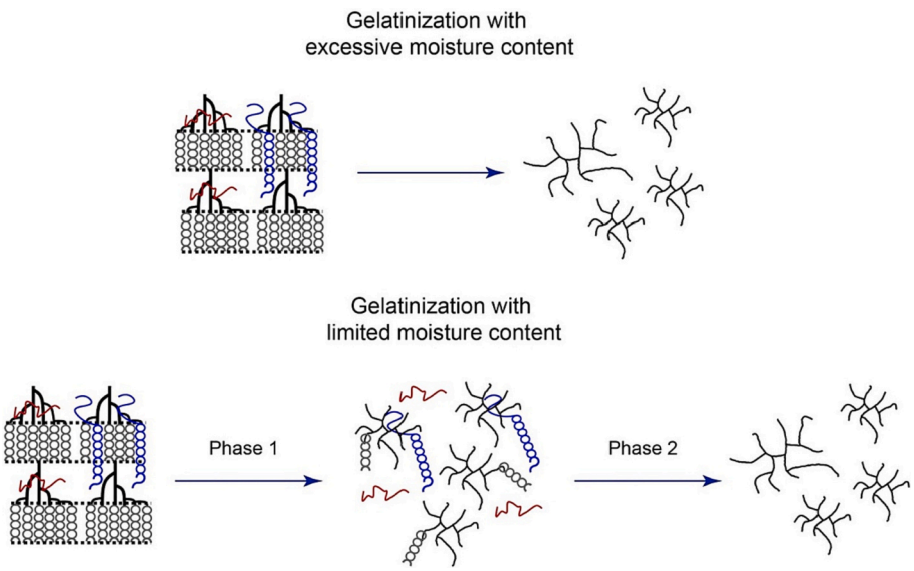

Fig. 3. Possible arrangement of starch molecules in the semi-crystalline lamella, accounting for the biphasic endothermic peaks when starch is gelatinized under limited moisture content. Red curves are amylose short chains; blue double helices are formed between amylopectin and amylose intermediate chains; black double helices are developed among amylopectin molecules. This graph is modified from the literature with permission (Li et al., 2022). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Effects of addition of salts, sugars, protein, lipids and non-starch polysaccharides on starch gelatinization property and possible mechanisms.

Ingredients	Effects on starch gelatinization	Possible mechanism
Salts	Depending on the types and concentrations, salts can cause either a depression or elevation of starch gelatinization temperature and enthalpy	(1) Structure-breaking and structure-making effects of salts on water; (2) electrostatic interactions between hydroxyl groups of starch and salts
Sugar	Sugars can generally increase starch gelatinization temperatures and alter gelatinization enthalpy	(1) Change of water activity; (2) change of glass transition temperatures; (3) hydrogen bonding interactions between sugars and starch
Protein	Protein can generally delay starch gelatinization	(1) Protein forms a physical matrix, restricting starch to swell; (2) protein competes with starch for free water; (3) protein acts as structural components (e.g. granule-channel proteins) in terms of maintaining the starch granule integrity
Lipids	Lipids can generally inhibit starch granule swelling (amylose leaching) and increase starch gelatinization temperatures	(1) Lipids form a layer on starch granules acting as a barrier against moisture distribution; (2) lipids interact with amylose molecules into amylose-lipid complexes
Non-starch polysaccharides	Non-starch polysaccharides can generally delay starch gelatinization	(1) Interacting with starch molecules; (2) inhibiting amylose leaching; (3) competing with the free water; (4) decreasing the plasticization capacity of the solvent; (5) forming physical barriers and inhibiting starch granule swelling

with potato starch, gelatinization peak temperature and enthalpy were decreased (Zhou et al., 2014). However, the gelatinization onset temperature and enthalpy were increased when the concentration of NaCl increased at the level of <4 mol/L (Jane, 1993). The gelatinization peak temperature of sago starch rose with NaCl up to 2 mol/L and then decreased, and gelatinization enthalpy only slightly increased at very low salt concentrations followed by a decline above a certain concentration (Ahmad & Williams, 1999). In terms of potato starch, addition of salting-out ions can increase the gelatinization temperature and enthalpy, while addition of slating-in ions can decrease these parameters (Zhou et al., 2014).

Various explanations have been proposed for explaining the effects of neutral salts on starch gelatinization during the last few decades (Table 2). The most popular ones are (1) structure-breaking and structure-making effects of salts on water and (2) electrostatic interactions between hydroxyl groups of starch and salts (Ahmad & Williams, 1999; Chiotelli et al., 2002; Jane, 1993; Li, Zhang, Ye, & Gao, 2015; Wang et al., 2017). These mechanisms were based on the model of liquid water structure, which was proposed as a mixture of hydrogen-bonded clusters and unbonded free-water molecules (Jane, 1993). The presence of free water could promote the interaction between starch and water through hydrogen bonds, which can replace the hydrogen bonds between starch chains, thus facilitating the dissociation of starch double helices during gelatinization. On the other hand, hydrogen-bonded clusters of water are less efficient in terms of breaking starch double helices. According to this model, ions with high charge density including Ca^{2+} and SO_4^{2-} have strong electrostatic interactions with water molecules (structure makers), which can reduce the fraction of free water and increase the viscosity of the solution. As a result, starch gelatinization temperature is increased. On the other hand, ions of low charge densities such as I^-, SCN^- and IO_4^- are known as "structure breakers", which can break or weaken hydrogen bonds between water molecules, decrease the viscosity of solution and increase the fraction of free water. In addition, ions with low charge densities tend to form helical complexes with starch molecules, which can further destabilize starch granules (Manca, Woortman, Loos, & Loi, 2015). Therefore, these salts can decrease the starch gelatinization temperature. Furthermore, considering the electronegative nature of starch molecules, anions generally tend to repel starch hydroxyl groups with the repulsion force proportional to the charge density of anions, which can stabilize starch granules. Cations can attract starch hydroxyl groups with the attraction force proportional to their charge density, which can destabilize starch granules (Ahmad & Williams, 1999). Therefore, for anions, an increase of charge density and concentration can cause a higher repulsion to starch hydroxyl groups and more structured water, which can both stabilize starch granules and increase starch gelatinization temperature. In terms of cations, an increased charge density can cause more structured water, which stabilizes starch granules. Whereas, a high concentration of these cations can attract more starch hydroxyl groups and

generate more heat, which destabilizes starch granules. The effects of cations on starch gelatinization property is thus more complex and depend on these two effects mentioned above, which are conflict with one another.

Although this model is helpful to explain many results as mentioned above, the understanding how salts alter starch gelatinization at the molecular level remains elusive. For example, in addition to the consensus on the existence of these two driving factors, there is little agreement on their relative importance and causes. Combined effects of different salts, moisture content and starch molecular structures on starch gelatinization property are also less clear. Therefore, much more efforts are still needed in the future to better understand the effects of salts on starch gelatinization property.

6. Effects of sugars on starch gelatinization

Sugars including natural sugar or sweeteners is another ingredient in diet, which can significantly change starch gelatinization property and food property. Especially, the widespread interest in starch gelatinization as a crucial starch functionality parameter in carbohydrate-based foods, as well as hindrance for reformulation efforts such as reducing added sugars, have led to a significant amount of efforts in understanding the effects of sugars on starch gelatinization property.

It is generally known that sugars increase starch gelatinization temperatures and alter gelatinization enthalpy, without affecting the mechanism of starch gelatinization in water (Table 2) (Perry & Donald, 2002; Renzetti et al., 2021). Three main hypotheses have been proposed as the underlying mechanisms in the literature, including (1) the change of water activity, (2) glass transition temperatures and (3) hydrogen bonding interactions between sugars and starch (Allan, Rajwa, & Mauer, 2018; Baek, Yoo, & Lim, 2004; Evans & Haisman, 1982; Slade & Levine, 1988). For example, sugar can compete water against starch and decreasing the water activity due to its hydrophilicity, which will finally increase starch gelatinization temperature (Evans & Haisman, 1982).

The effects of sugars on starch gelatinization depend on the amount, molecular weight and intrinsic properties of the sugars (Allan et al., 2018; Baek et al., 2004; Perry & Donald, 2002). For example, sucrose was shown to increase starch gelatinization temperatures more than other sugars (Allan et al., 2018). This has developed challenges in terms of replacing sucrose functionality in products such as baked goods, wherein monitor of starch gelatinization is a critical quality parameter (van der Sman & Renzetti, 2018). Another study has compared the effects of 19 different low molecular weight carbohydrates on wheat starch gelatinization temperatures, and concluded that sweeteners interact with starch in the amorphous regions, which restrict the mobility of amorphous regions and increase its gelatinization temperatures to overcome these attractive forces (Allan et al., 2018). An increase in the sweetener concentration in the solution would increase the amount of sweetener entering starch granules, subsequently increasing the wheat starch gelatinization temperature (Allan et al., 2018). There was also a positive correlation between the molecular weight of a sweetener and the starch gelatinization temperature (Allan et al., 2018). Among different properties of sweeteners, the viscosity was strongest factor in determining the starch gelatinization (Allan et al., 2018).

Some new insights have also been learnt recently on the effects of sugars on starch gelatinization property. Water, sugars and polyols have in common that they interact with biopolymers such as starch via hydrogen bonding. Their mixtures might thus be treated as a single effective solvent, as the nature of the specific hydrogen bonding interactions is same (Sman, 2016). This will largely simplify the analysis of effects of sugars on starch gelatinization. It has been recently proposed that the effects of sugars or sugar replacers on starch gelatinization property can be predicted by the volumetric density of hydrogen bonds, which are available for intermolecular hydrogen bonds (Sman & Mauer, 2019). Starch gelatinization temperatures including onset, peak and conclusion temperatures were shown to be a function of the volume density of available hydrogen bonding sites in solutions containing sugars, polyols, amino acids or oligofructoses for interacting with starch (Renzetti et al., 2021). In this theory, starch swelling (caused by solvent ingress and amylopectin helix-helix dissociation) is affected by kinetic factors associated with the viscosity and size of plasticizers, which can be described by the molar volume density of effective hydroxyl groups of plasticizers (an intrinsic property of the plasticizer), as well as thermodynamic factors affected by sugar partitioning and hydrogen bonding ability (both related to volumetric density of hydrogen bonds in sugar solutions treated as a single solvent); further melting of starch crystalline regions due to the helix-coil transition is determined by thermodynamics depending on solvent hydrogen bonding ability (Renzetti et al., 2021). The amount of intermolecular interactions between starch and sugars were shown to be related to the sugar concentration, stereochemistry and molecular flexibility (Allan et al., 2018). For example, sugars with the same number of carbons and molar concentrations could have varying effects on starch gelatinization temperatures if they have different stereochemistry (Allan et al., 2018).

7. Effects of protein on starch gelatinization

Protein is another critical component in many foods, and a better understanding of starch-protein interaction and its effects on starch gelatinization property thus has a practical application in terms of developing functional foods with desirable properties. Endogenous protein bodies can form a matrix surrounding starch granules in adlay seed flour, inhibiting starch gelatinization; removal of these endogenous proteins increases starch welling power and gelatinization enthalpy while decreases gelatinization temperatures (Ding et al., 2021). Similarly, Hamaker & Griffin has found that protein with disulfide bonds can inhibit the swelling of starch granules from rice flour during gelatinization (Hamaker & Griffin, 1993). In addition, addition of endogenous protein back to rice starch has increased the starch gelatinization temperature, possibly due to its competition with starch on absorbing free water (Wang, Zhang, Wang, Ai, & Xiong, 2020). Removal both granule-channel and granule-associated proteins increased amylose leaching and swelling power, reduced the gelatinization temperature, while had no significant effects on the gelatinization enthalpy of rice starch, possibly due to the functionality of granule-associated proteins in maintaining starch structural stability (Zhan et al., 2019). Except for endogenous protein, addition of exogenous proteins could also largely affect starch gelatinization property. For example, the swelling of corn starch granules was restricted by the addition of whey protein isolate, possibly due to the restriction of water availability (Yang, Zhong, Douglas Goff, & Li, 2019).

To sum, different mechanisms exist in the literature in terms of explaining the effects of either endogenous or exogenous protein on starch gelatinization property (Table 2). They can be generally grouped as (1) protein forms a physical matrix, restricting starch to swell; (2) protein competes with starch for free water; (3) protein acts as structural components (e.g. granule-channel proteins) in terms of maintaining the starch granule integrity.

8. Effects of lipids on starch gelatinization

Lipid is an important food ingredient, whereas its effects on starch gelatinization are complex and controversial findings could be found in the literature. Generally, addition of lipids can inhibit starch granule swelling (amylose leaching) and increase starch gelatinization temperatures by acting as a barrier against moisture distribution or forming complexes with amylose molecules (Table 2) (Dun et al., 2020; Gerits, Pareyt, & Delcour, 2015; Wang, Wang, Yu, & Wang, 2016; Yu, Wang, Chen, Li, & Wang, 2018). It has been shown that for a low apparent amylose content mutant (GZ93) and parent rice (R7954), removal of internal starch lipids has decreased the gelatinization peak temperature and enthalpy while significantly increased the swelling powder (Zhang

et al., 2019). However, for a high-lipid rice mutant (GZ1), the removal of internal starch lipids has increased the gelatinization enthalpy (Zhang et al., 2019). Removal of endogenous lipids has also increased the gelatinization enthalpy of adlay seed starch (Ding et al., 2021). It suggests that the effects of lipids on starch gelatinization depend on the starch sources. In addition, lipids in grains can be generally categorized into starch lipids (including both internal starch lipids and starch surface lipids) and non-starch lipids (Abdel-Aal, Hernandez, Rabalski, & Hucl, 2020). Internal starch lipids are an integral part of starch granules as they can form inclusion complexes with amylose, while starch surface lipids are loosely associated with the surface layers of the granules and possibly derived from the non-starch lipids. Starch internal and surface lipids and their fatty acids were significantly correlated with most of starch gelatinization and pasting properties (Abdel-Aal et al., 2020). It remains to be further studied with respects to the mechanisms how different lipid components (e.g. internal vs surface) determine starch gelatinization property.

9. Effects of non-starch polysaccharides on starch gelatinization

Interactions between starch and other food ingredients such as non-starch polysaccharides during food processing could also largely affect starch properties such as gelatinization (Donmez et al., 2021; Tu, Brennan, & Brennan, 2021). Non-starch polysaccharides have been frequently included in starch-based foods in order to improve starch physiochemical properties, as native starch is frequently shearing and heating instable and acidic sensitive (Liu et al., 2022; Luo et al., 2021). For example, tamarind seed polysaccharide can bind with starch granules and change the starch molecular conformation, which finally retard starch gelatinization (Xie, Zhang, Xia, & Ai, 2020). Hydrocolloids such as xanthan and sodium alginate can form a hydration film on the surface of maize starch granules via intermolecular hydrogen bonding cross-linking or coating (Fig. 4), which could subsequently increase the water holding capacity while retarding the gelatinization of maize starch granules (Lee, Lee, Chung, & Park, 2021). Non-starch polysaccharides with different structures have distinct effects on starch gelatinization property; water-extractable arabinoxylan with low molecular weight can hinder starch gelatinization more evidently than arabinoxylan with high molecular weight, possibly due to the pronounced inhibition on amylose leaching (Hou et al., 2020). Similarly, corn fiber gum (a novel arabinoxylan hydrocolloid) can interact with leached amylose through intermolecular hydrogen bonds or entanglements, stabilizing the paste system (Qiu et al., 2016). Rhamnogalacturonan type I (RG-I) can also adsorb on potato starch granule surface, inhibiting granule swelling and amylose leaching, which finally delay potato starch gelatinization (Yin et al., 2021). Inulin and *Mesona chinensis* polysaccharide (an acid polysaccharide extracted from *M. chinensis*) could increase wheat and maize starch gelatinization temperature possibly due to the reduced water availability (Luo et al., 2017; Luo et al., 2021). Effects of *Mesona chinensis* polysaccharide on starch gelatinization might also depend on its extraction methodology, which could largely determine the polysaccharide structure (Xiao et al., 2020). For example, amylose leaching in tapioca starch was inhibited by adding *Mesona chinensis* polysaccharide obtained by cellulase-assisted extraction, while it was promoted by adding polysaccharide obtained by sodium carbonate-assisted extraction method (Xiao et al., 2020). Onset and peak gelatinization temperatures of wheat starch were positively affected by the bran concentration in water possibly due to that water-extractable bran components decreased the plasticization capacity of the solvent (De Bondt, Liberloo, Roye, Goos, & Courtin, 2020). The presence of intact cell wall could exert a significant retarding effect on starch gelatinization, through impeding heat transfer, restricting starch granule swelling and limiting water ingress (Guo, Yu, Wang, & Wang, 2020).

In summary, non-starch polysaccharides could affect starch gelatinization through (1) interacting with starch molecules; (2) inhibiting amylose leaching; (3) competing with the free water; (4) decreasing the plasticization capacity of the solvent; and (5) forming physical barriers and inhibiting starch granule swelling (Table 2). The different mechanisms are depending on the structural difference of these non-starch polysaccharides.

10. Conclusions and future remarks

New insights on the effects of different factors including starch molecular structure, moisture content, salts, sugars, protein, lipids and non-starch polysaccharides on starch gelatinization property have been learnt recently. Especially, kinetics modeling has been developed for the starch gelatinization process under both limited and excessive water content. Amylose with different chain lengths can have distinct effects on the starch gelatinization temperatures and enthalpy by interacting with amylopectin chains in the semi-crystalline lamella. Moisture is a crucial factor in determining starch gelatinization property, with new evidence suggesting that different groups of amylopectin-amylopectin or amylose-amylopectin double helices are accounting for the observed multi-endothermic peaks from the starch gelatinization process under limited moisture content. The presence of salts, sugars,

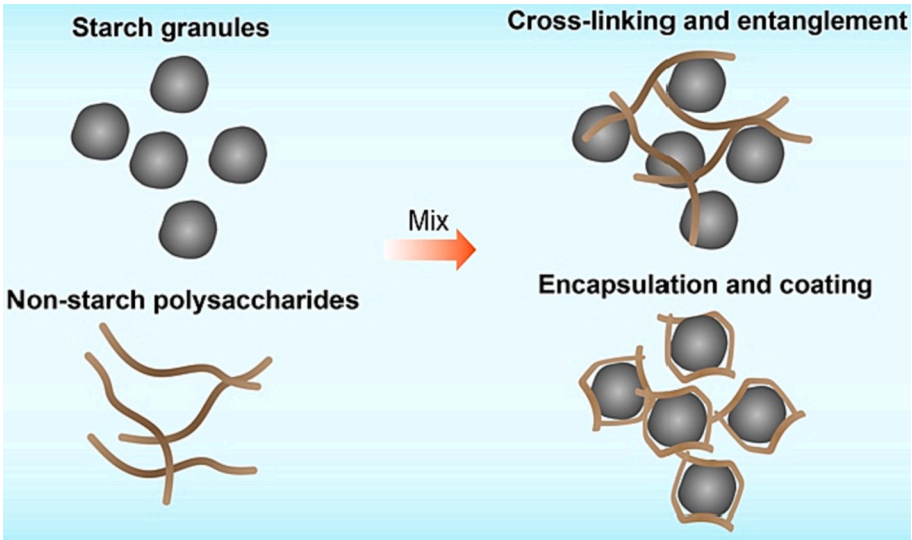

Fig. 4. Formation of a hydration film on the surface of starch granules via cross-linking (entanglement) or encapsulation (coating) by non-starch polysaccharides.

protein, lipids and non-starch polysaccharides can determine starch gelatinization process, through various mechanisms, which were all summarized in the current review.

However, despite many progresses have been achieved during last few decades in terms of understanding starch gelatinization property, many essential questions remain to be answered in the future. For example, although it is clear that different food ingredients could affect starch gelatinization property, it is less understood if and how these different ingredients combinedly determine starch gelatinization property. Most of past studies have only focused on the investigation of a single or a few of these ingredients and their effects on starch gelatinization property. Therefore, it is still needed to systematically investigate effects of all these different ingredients on starch gelatinization property in the future. The inclusion of various gelatinization conditions such as moisture content as another dimension with these ingredients is also required to fully understand starch gelatinization property.

Considering the importance of starch gelatinization process in terms of determining the texture and nutrition property of starch-based foods, this review contains important information which can act as basis for the development of new generations of functional foods with desirable properties in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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